

Introduction to **Information Retrieval**

Evaluation

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CS276 – Information Retrieval and Web Search

Revised by

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How do you tell if users are happy?

- Search returns products relevant to users
 - How do you assess this at scale?
- Search results get clicked a lot
 - Misleading titles/summaries can cause users to click
- Users buy after using the search engine
 - Or, users spend a lot of \$ after using the search engine
- Repeat visitors/buyers
 - Do users leave soon after searching?
 - Do they come back within a week/month/... ?

Happiness: elusive to measure

- Most common proxy: relevance of search results
 - Pioneered by Cyril Cleverdon in the Cranfield Experiments



- But how do you measure relevance?

Measuring relevance

- Three elements:
 1. A benchmark document collection
 2. A benchmark suite of queries
 3. An assessment of either Relevant or Nonrelevant for each query and each document

Early public test Collections (20th C)

TABLE 4.3 Common Test Corpora

<i>Collection</i>	<i>NDocs</i>	<i>NQrys</i>	<i>Size (MB)</i>	<i>Term/Doc</i>	<i>Q-D RelAss</i>
ADI	82	35			
AIT	2109	14	2	400	>10,000
CACM	3204	64	2	24.5	
CISI	1460	112	2	46.5	
Cranfield	1400	225	2	53.1	
LISA	5872	35	3		
Medline	1033	30	1		
NPL	11,429	93	3		
OSHMED	34,8566	106	400	250	16,140
Reuters	21,578	672	28	131	
TREC	740,000	200	2000	89-3543	» 100,000

Typical
TREC

Recent datasets: 100s of million web pages (GOV, ClueWeb, ...)

Evaluating an IR system

- Note: **user need** is translated into a **query**
- Relevance is assessed relative to the **user need**, *not* the **query**
- E.g., Information need: *My swimming pool bottom is becoming black and needs to be cleaned.*
- Query: ***pool cleaner***
- Assess whether the doc addresses the underlying need, not whether it has these words

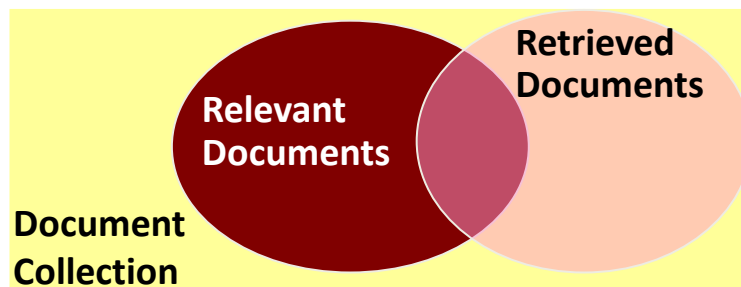
Motivations

- Among different IR systems/models/algorithms which one is the best ?
- What's the best component for :
 - ranking (inner product, cosine, ...)
 - Term Selection (stopword, stemming...)
 - Term weighting (TF, TF-IDF,...)
- When to stop (cut) the ranked list of retrieved documents?

Labeled Document Collections (Gold Standard)

- Given a target Document Collection:
- STEP 1. Develop a set of representative queries.
- STEP 2. Use a basic IR technology with wide coverage (see later about recall) to gather **all** relevant candidate document
- STEP 3. For each query, one or more experts establish the relevance of the different documents of the collection selected at STEP 2.
 - Typically the decision is categorical (i.e. binary)
- The overall process is quite costly in terms of involved human resources as document collections should be representative (i.e. very large)

Precision & Recall



relevant irrelevant	retrieved & irrelevant	not retrieved & irrelevant
	retrieved & relevant	not retrieved & relevant
	returned	Not returned

$$\text{recall} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents}}$$

$$\text{precision} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of documents retrieved}}$$

Precision & Recall

relevant	tp (true positive)	fn (false negative)
irrelevant	fp (false positive)	tn (true negative)
	retrieved	not retrieved

$$recall = \frac{tp}{tp + fn}$$

$$precision = \frac{tp}{tp + fp}$$

Accuracy and Error rate

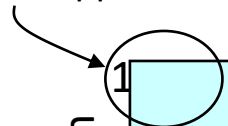
relevant	tp (true positive)	fn (false negative)
	fp (false positive)	tn (true negative)
irrelevant	retrieved	not retrieved

$$acc = \frac{tp + tn}{tp + tn + fn + fp}$$

$$err = \frac{fp + fn}{tp + tn + fn + fp}$$

Recall vs. Precision

Restituisce documenti rilevanti
ma ne perde troppi utili

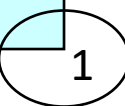


Precision

0

Recall

Prestazione Ideale



Restituisce per la maggior
parte dei documenti rilevanti
ma ne include troppi inutili

Evaluation Metrics: Examples

 = the relevant documents

Ranking #1

										
Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2

										
Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

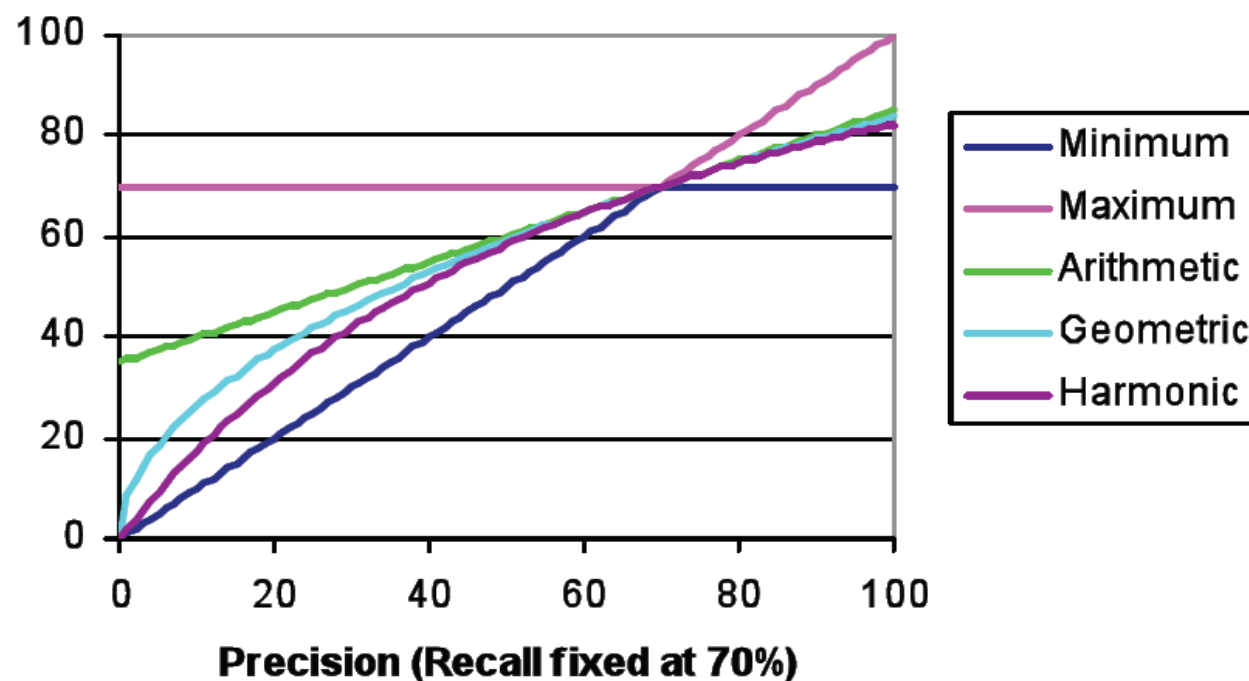
F Measure

- Harmonic mean of recall and precision

$$F = \frac{1}{\frac{1}{2}(\frac{1}{R} + \frac{1}{P})} = \frac{2RP}{(R+P)}$$

- Why harmonic mean?
- harmonic mean emphasizes the importance of small values, whereas the arithmetic mean is affected more by outliers that are unusually large

Different Combinations of *precision & recall*



Rank-Based Measures

- Binary relevance
 - Precision@K ($P@K$)
 - Mean Average Precision (MAP)
 - Mean Reciprocal Rank (MRR)
- Multiple levels of relevance
 - Normalized Discounted Cumulative Gain (NDCG)

Precision@K

- Set a rank threshold K
- Compute % relevant in top K
- Ignores documents ranked lower than K

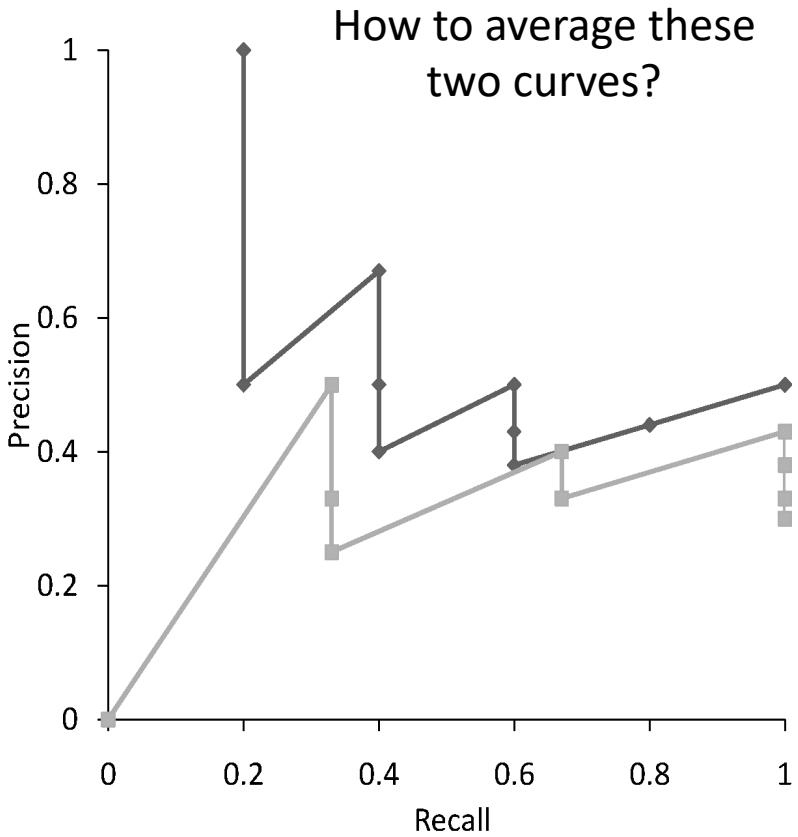
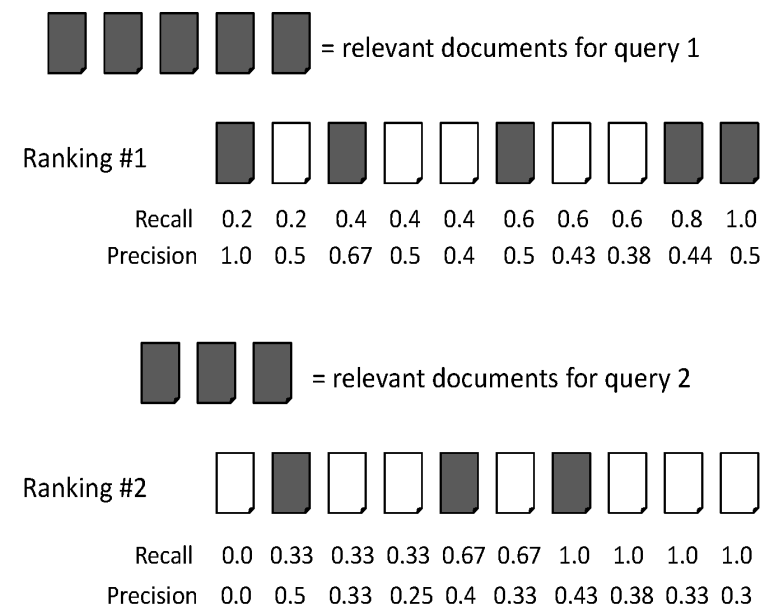
■ Ex:

- $\text{Prec}@3$ of $2/3$
- $\text{Prec}@4$ of $2/4$
- $\text{Prec}@5$ of $3/5$



- In similar fashion we have $\text{Recall}@K$

Recall-Precision Graph

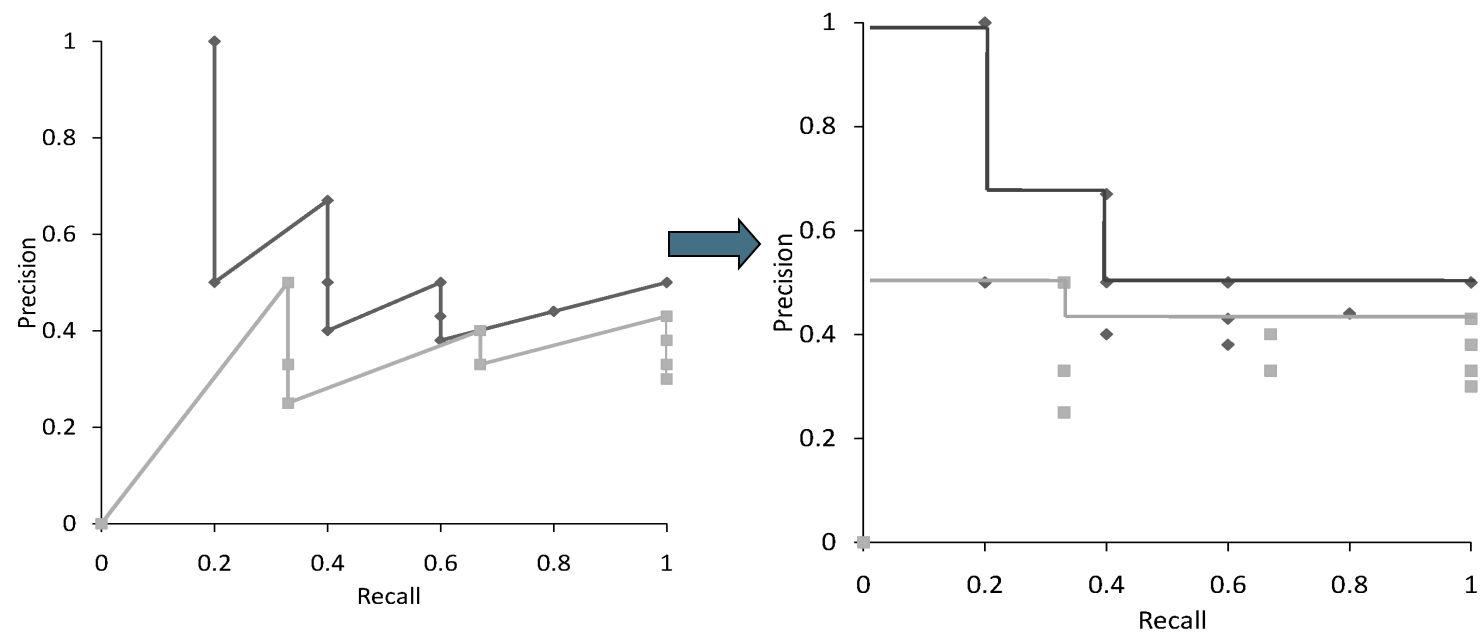


Interpolation

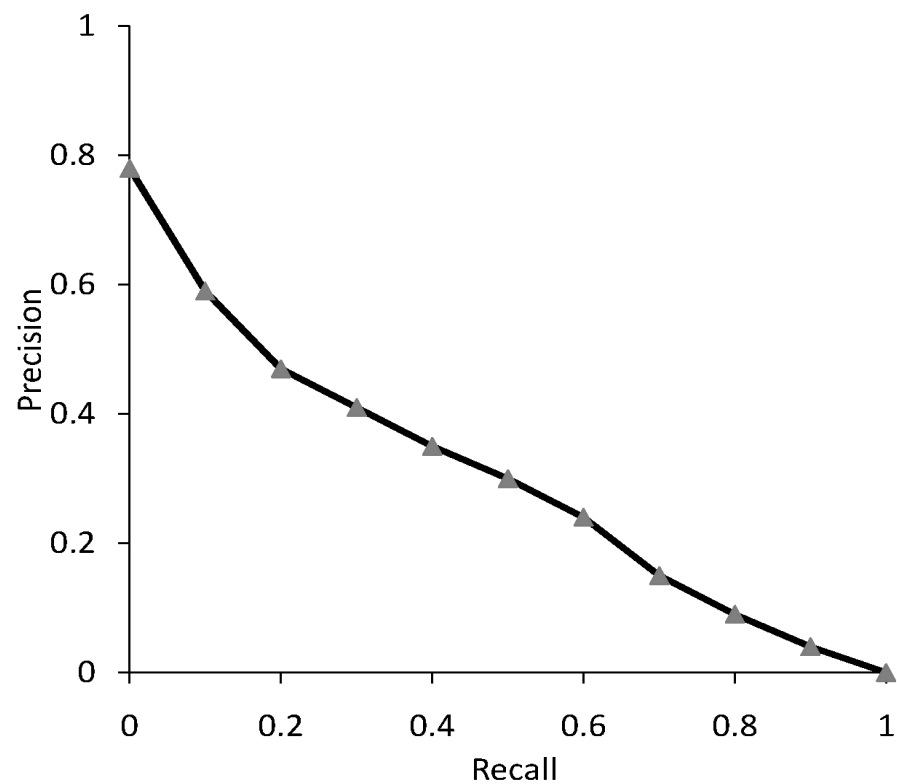
$$P(R) = \max\{P' : R' \geq R \wedge (R', P') \in S\}$$

- where S is the set of observed (R, P) points
- Defines precision at any recall level as the maximum precision observed in any recall-precision point at a higher recall level
 - produces a step function
 - defines precision at recall 0.0

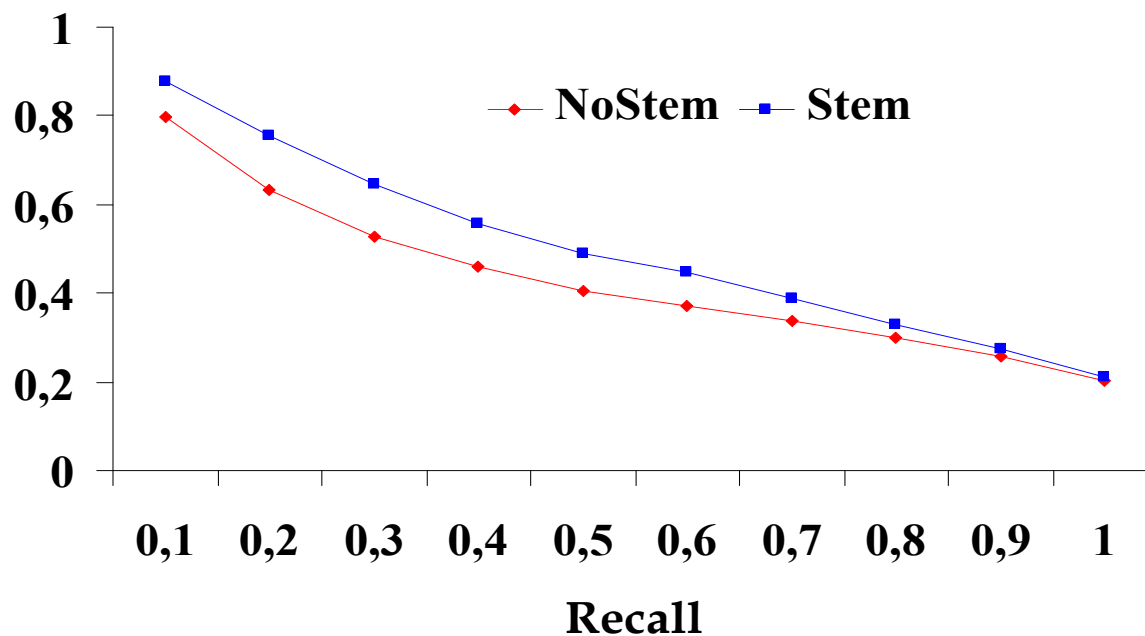
Interpolation



Graph for 50 Queries

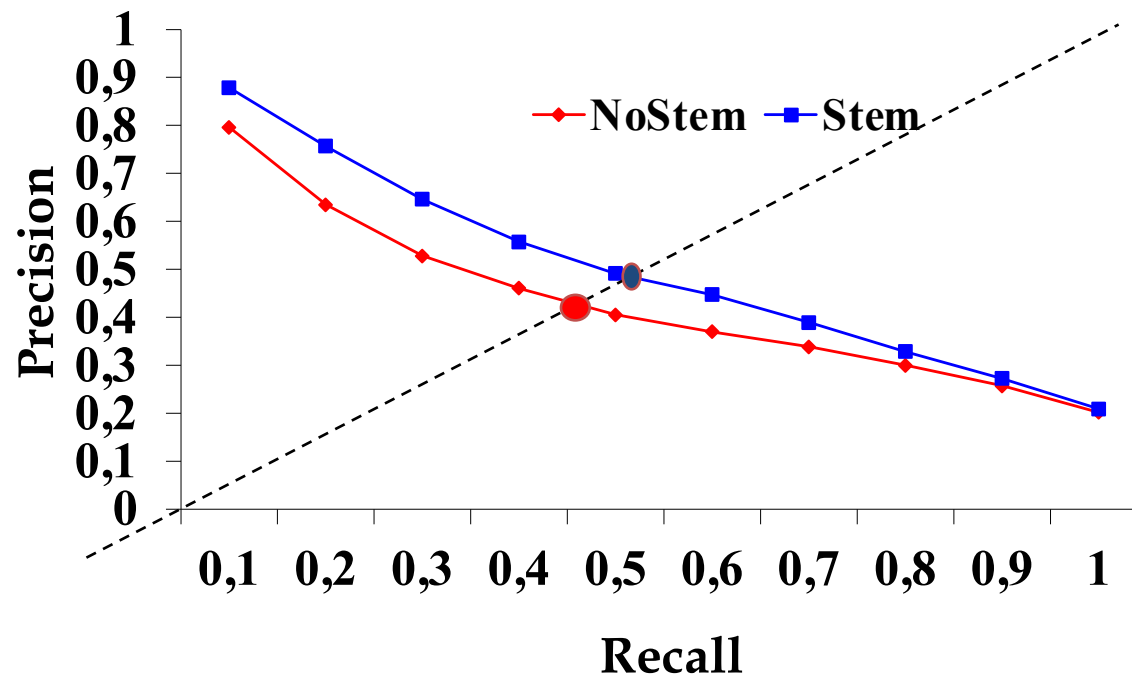


Comparison among different IR Systems



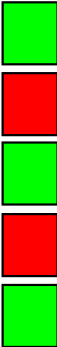
Breakeven point

Is the interpolated value for which *precision equals recall*



Mean Average Precision

- Consider rank position of each **relevant** doc
 - $K_1, K_2, \dots K_R$
- Compute Precision@K for each $K_1, K_2, \dots K_R$
- Average precision = average of P@K

- Ex:  has AvgPrec of $\frac{1}{3} \cdot \left(\frac{1}{1} + \frac{2}{3} + \frac{3}{5} \right) \approx 0.76$

- MAP is Average Precision across multiple queries/rankings

Average Precision

 = the relevant documents

Ranking #1

										
Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2

										
Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

Ranking #1: $(1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$

Ranking #2: $(0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$

MAP

 = relevant documents for query 1

Ranking #1

										
Recall	0.2	0.2	0.4	0.4	0.4	0.6	0.6	0.6	0.8	1.0
Precision	1.0	0.5	0.67	0.5	0.4	0.5	0.43	0.38	0.44	0.5

 = relevant documents for query 2

Ranking #2

										
Recall	0.0	0.33	0.33	0.33	0.67	0.67	1.0	1.0	1.0	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.33	0.43	0.38	0.33	0.3

$$\text{average precision query 1} = (1.0 + 0.67 + 0.5 + 0.44 + 0.5)/5 = 0.62$$

$$\text{average precision query 2} = (0.5 + 0.4 + 0.43)/3 = 0.44$$

$$\text{mean average precision} = (0.62 + 0.44)/2 = 0.53$$

Mean average precision

- If a relevant document never gets retrieved, we assume the precision corresponding to that relevant doc to be zero
- MAP is macro-averaging: each query counts equally
- Now perhaps most commonly used measure in research papers
- Good for web search?
- MAP assumes user is interested in finding many relevant documents for each query
- MAP requires many relevance judgments in text collection

BEYOND BINARY RELEVANCE



Web Images Video Local Shopping More ▾

Toyota safety

Search

Options ▾

 Search Pad

 SearchScan - On

108,000,000 results for **Toyota safety**:

Show All

 Toyota

 Motor Trend

 CarsDirect

 Shopping Sites

Also try: [toyota safety ratings](#), [toyota safety recall](#), [More...](#)

Toyota Recall

Toyota Takes Care of its Customers. Read the FAQs at **Toyota.com**.
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Toyota home page for car **safety** and car technology Prius model.
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We are presenting **Toyota's safety** technologies for cars. We clearly explain about car **safety** and car technology using movies and more.
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Toyota Safety Ratings - Toyota Safety Features - Motor Trend ...

MotorTrend offers **Toyota safety** ratings, comprehensive auto **safety** reports, and more. View a all of the standard **Toyota safety** features. ...
[motortrend.com/new_cars/07/toyota/safety_ratings/index.html](#) - 149k - [Cached](#)

Toyota Motor Europe Corporate Site Safety

Our approach. **Toyota** believes that all stakeholders in the road **safety** equation share a responsibility to reduce the frequency of road accidents. ...
[www.toyota.eu/Safety](#) - [Cached](#)

pdf European Safety Brochure 2005

4047k - Adobe PDF - [View as html](#)
not guarantee that all accidents or injuries will be avoided when driving a **Toyota** and/or Lexus brand motor vehicle equipped with the **safety** systems ...
[www.toyota.no/Images/Safety_Brochure_tcm308-344461.pdf](#)

Toyota - Star Safety System

Star **Safety** System ... **Toyota** Mobility Program. Careers. Contact Us. Home. contact us. site map. your privacy rights. legal terms. **Toyota** Newsroom. sign up for info ...
[www.toyota.com/vehicles/demos/star-safety.html](#) - 58k - [Cached](#)

Toyota Prius Safety Ratings - CarsDirect

Get overall **safety** ratings and NHTSA crash test results for the **Toyota** Prius at CarsDirect.

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Safety Toyota
Explore 5,000+ Pro Sports Choices. Save On Safety Toyota.
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[See your message here...](#)

Sponsored Results

Toyota Recall
Sponsored Results

fair

fair

Good

Discounted Cumulative Gain

- Popular measure for evaluating web search and related tasks
- Two assumptions:
 - Highly relevant documents are more useful than marginally relevant documents
 - the lower the ranked position of a relevant document, the less useful it is for the user, since it is less likely to be examined

Discounted Cumulative Gain

- Uses *graded relevance* as a measure of usefulness, or *gain*, from examining a document
- Gain is accumulated starting at the top of the ranking and may be reduced, or *discounted*, at lower ranks
- Typical discount is $1/\log(\text{rank})$
 - With base 2, the discount at rank 4 is $1/2$, and at rank 8 it is $1/3$

Summarize a Ranking: DCG

- What if relevance judgments are in a scale of $[0, r]$? $r > 2$
- Cumulative Gain (CG) at rank n
 - Let the ratings of the n documents be $r_1, r_2, \dots, r_n$ (in ranked order)
 - $CG = r_1 + r_2 + \dots + r_n$
- Discounted Cumulative Gain (DCG) at rank n
 - $DCG = r_1 + r_2 / \log_2 2 + r_3 / \log_2 3 + \dots + r_n / \log_2 n$
 - We may use any base for the logarithm

Discounted Cumulative Gain

- *DCG* is the total gain accumulated at a particular rank p :

$$DCG_p = rel_1 + \sum_{i=2}^p \frac{rel_i}{\log_2 i}$$

- Alternative formulation:

$$DCG_p = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log(1+i)}$$

- used by some web search companies
- emphasis on retrieving highly relevant documents

DCG Example

- 10 ranked documents judged on 0–3 relevance scale:
3, 2, 3, 0, 0, 1, 2, 2, 3, 0
- discounted gain:
 $3, 2/1, 3/1.59, 0, 0, 1/2.59, 2/2.81, 2/3, 3/3.17, 0$
 $= 3, 2, 1.89, 0, 0, 0.39, 0.71, 0.67, 0.95, 0$
- DCG:
3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61

NDCG for summarizing rankings

- Normalized Discounted Cumulative Gain (NDCG) at rank n
 - Normalize DCG at rank n by the DCG value at rank n of the ideal ranking
 - The ideal ranking would first return the documents with the highest relevance level, then the next highest relevance level, etc
- Normalization useful for contrasting queries with varying numbers of relevant results
- NDCG is now quite popular in evaluating Web search

NDCG - Example

4 documents: d_1, d_2, d_3, d_4

i	Ground Truth		Ranking Function ₁		Ranking Function ₂	
	Document Order	r_i	Document Order	r_i	Document Order	r_i
1	d4	2	d3	2	d3	2
2	d3	2	d4	2	d2	1
3	d2	1	d2	1	d4	2
4	d1	0	d1	0	d1	0
	NDCG _{GT} =1.00		NDCG _{RF1} =1.00		NDCG _{RF2} =0.9203	

$$DCG_{GT} = 2 + \left(\frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.6309$$

$$DCG_{RF1} = 2 + \left(\frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.6309$$

$$DCG_{RF2} = 2 + \left(\frac{1}{\log_2 2} + \frac{2}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.2619$$

$$MaxDCG = DCG_{GT} = 4.6309$$

What if the results are not in a list?

- Suppose there's only one Relevant Document
- Scenarios:
 - known-item search
 - navigational queries
 - looking for a fact
- Search duration $\sim$ Rank of the answer
 - measures a user's effort

Mean Reciprocal Rank

- Consider rank position, K , of first relevant doc
 - Could be – only clicked doc
- Reciprocal Rank score = $\frac{1}{K}$
- MRR is the mean RR across multiple queries

Human judgments are

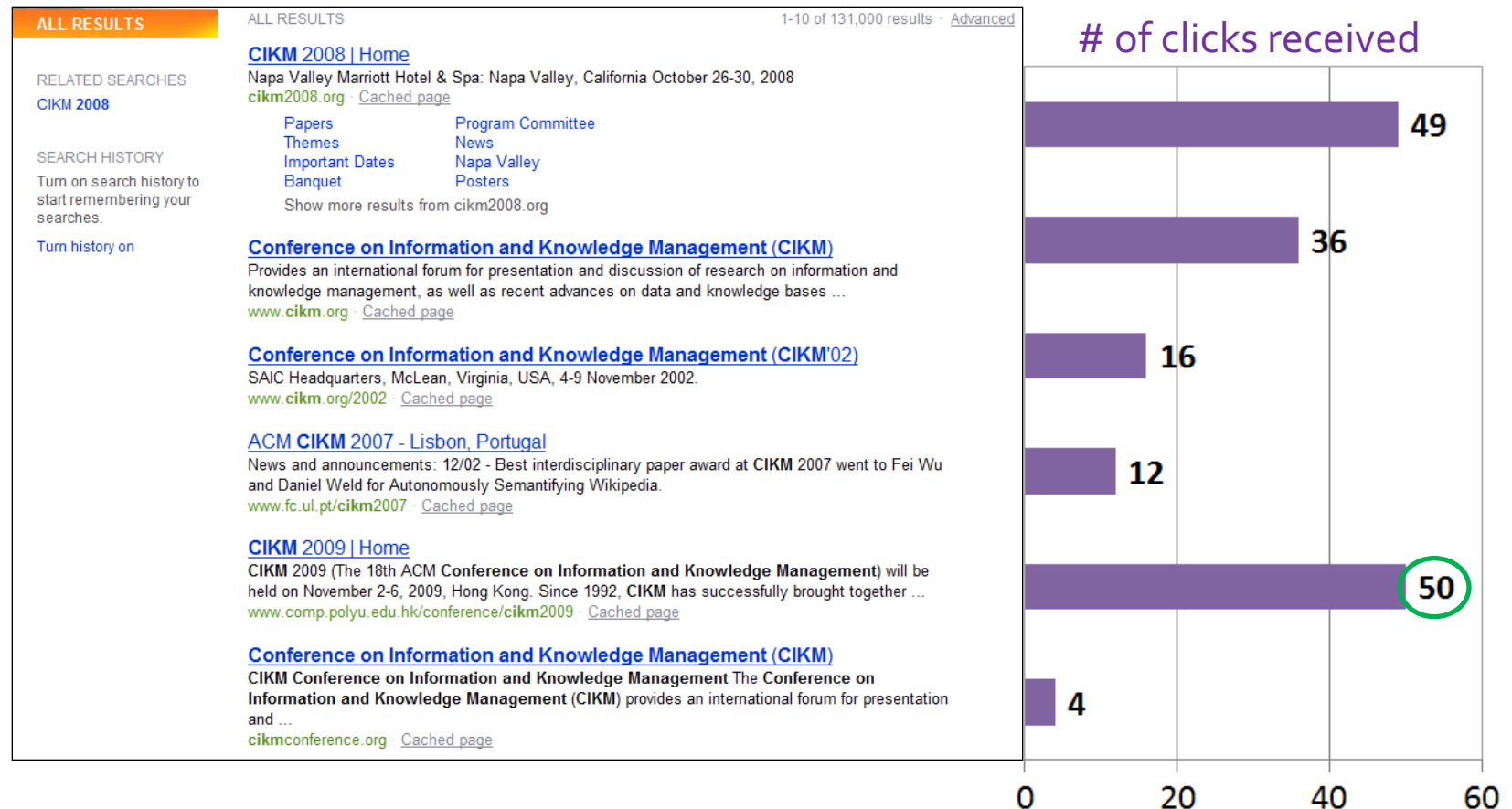
- Expensive
- Inconsistent
 - Between raters
 - Over time
- Decay in value as documents/query mix evolves
- Not always representative of “real users”
 - Rating vis-à-vis query, don’t know underlying need
 - May not understand meaning of terms, etc.
- So – what alternatives do we have?

USING USER CLICKS

User Behavior

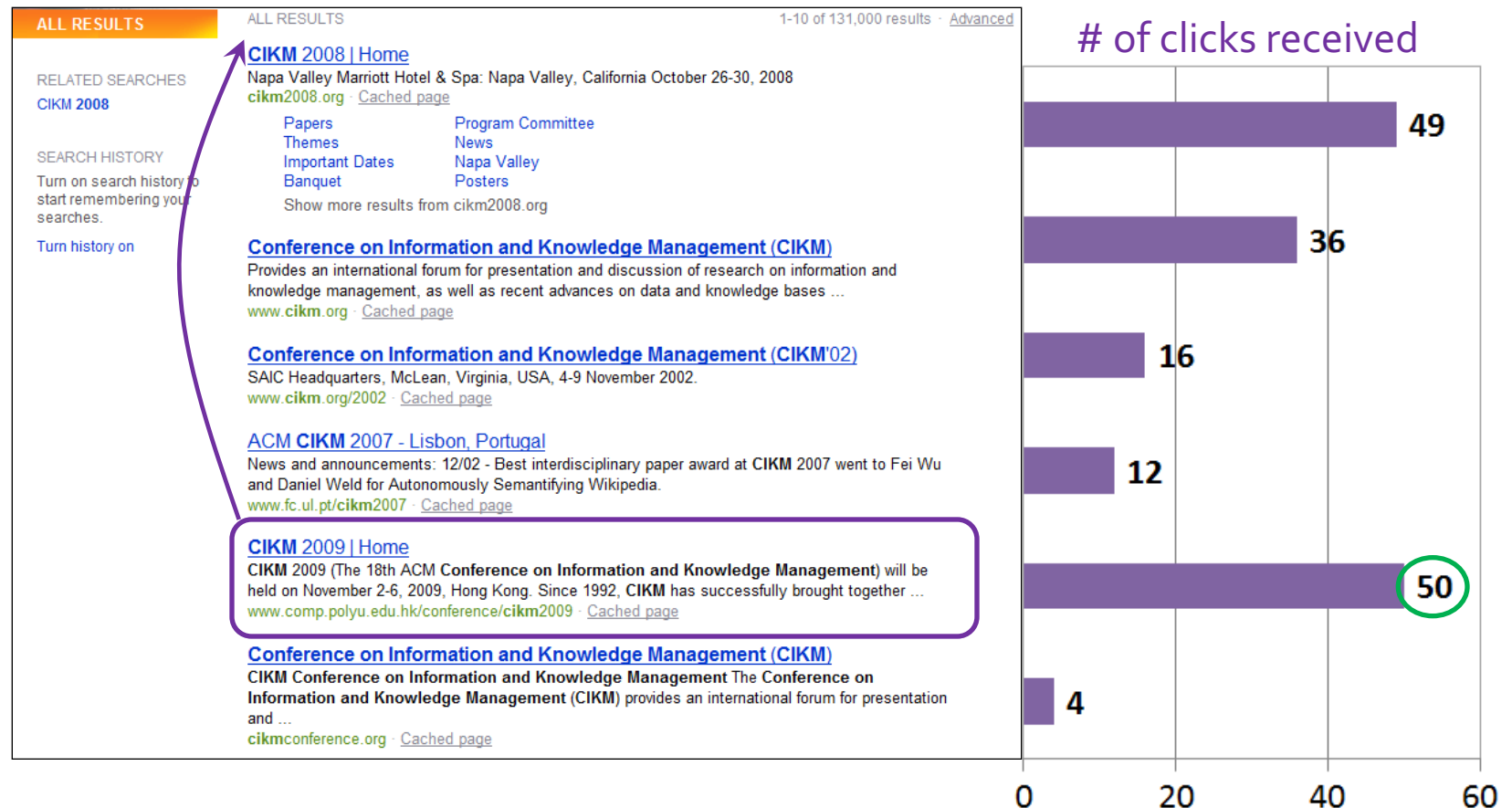
Taken with slight adaptation from Fan Guo and Chao Liu's 2009/2010 CIKM tutorial: Statistical Models for Web Search: Click Log Analysis

■ Search Results for "CIKM" (in 2009!)



User Behavior

■ Adapt ranking to user clicks?



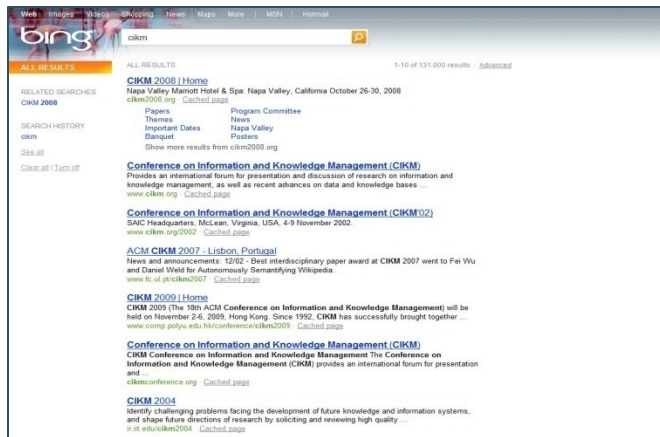
What do clicks tell us?

■ Tools needed for non-trivial cases

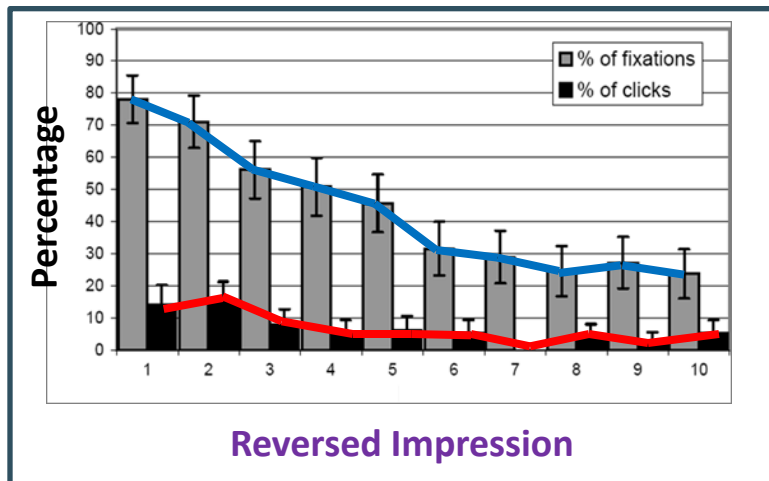
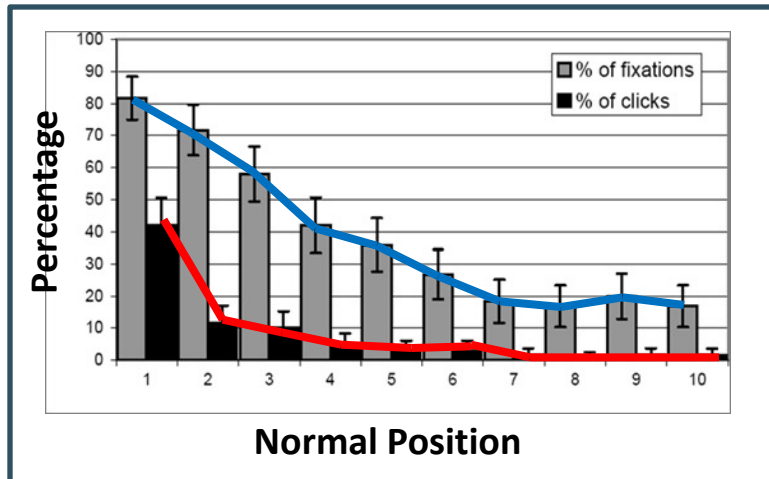


Strong position bias, so absolute click rates unreliable

Eye-tracking User Study



Click Position-bias



- Higher positions receive more **user attention (eye fixation)** and **clicks** than lower positions.
- This is true even in the extreme setting where the order of positions is **reversed**.
- “Clicks are informative but biased”.

[Joachims+07]

Relative vs absolute ratings

The screenshot shows a search engine results page for the query 'CIKM 2008'. The page includes a sidebar with 'RELATED SEARCHES' (CIKM 2008) and 'SEARCH HISTORY'. The main results area shows four entries, each with a title, description, and URL. Three blue arrows indicate a user's click sequence: the first arrow points to the 'CIKM 2008 | Home' link, the second arrow points to the 'Conference on Information and Knowledge Management (CIKM'02)' link, and the third arrow points to the 'Conference on Information and Knowledge Management (CIKM)' link.

ALL RESULTS 1-10 of 131,000 results · [Advanced](#)

ALL RESULTS

RELATED SEARCHES
[CIKM 2008](#)

SEARCH HISTORY
Turn on search history to start remembering your searches.
[Turn history on](#)

[CIKM 2008 | Home](#)
Napa Valley Marriott Hotel & Spa: Napa Valley, California October 26-30, 2008
[cikm2008.org](#) · [Cached page](#)

Papers Program Committee
Themes News
Important Dates Napa Valley
Banquet Posters
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[Conference on Information and Knowledge Management \(CIKM\)](#)
Provides an international forum for presentation and discussion of research on information and knowledge management, as well as recent advances on data and knowledge bases ...
[www.cikm.org](#) · [Cached page](#)

[Conference on Information and Knowledge Management \(CIKM'02\)](#)
SAIC Headquarters, McLean, Virginia, USA, 4-9 November 2002.
[www.cikm.org/2002](#) · [Cached page](#)

[ACM CIKM 2007 - Lisbon, Portugal](#)
News and announcements: 12/02 - Best interdisciplinary paper award at CIKM 2007 went to Fei Wu and Daniel Weld for Autonomously Semantifying Wikipedia.
[www.fc.ul.pt/cikm2007](#) · [Cached page](#)

[CIKM 2009 | Home](#)
CIKM 2009 (The 18th ACM Conference on Information and Knowledge Management) will be held on November 2-6, 2009, Hong Kong. Since 1992, CIKM has successfully brought together ...
[www.comp.polyu.edu.hk/conference/cikm2009](#) · [Cached page](#)

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User's click sequence

Hard to conclude Result1 > Result3
Probably can conclude Result3 > Result2

Evaluating pairwise relative ratings

- Pairs of the form: DocA better than DocB for a query
 - Doesn't mean that DocA relevant to query
- Now, rather than assess a rank-ordering wrt per-doc relevance assessments ...
- Assess in terms of conformance with historical pairwise preferences recorded from user clicks
- BUT!
- Don't learn and test on the same ranking algorithm

Comparing two rankings via clicks (Joachims 2002)

Query: [support vector machines]

Ranking A

Kernel machines
SVM-light
Lucent SVM demo
Royal Holl. SVM
SVM software
SVM tutorial

Ranking B

Kernel machines
SVMs
Intro to SVMs
Archives of SVM
SVM-light
SVM software

Interleave the two rankings

This interleaving
starts with B

Kernel machines
Kernel machines
SVMs
SVM-light
Intro to SVMs
Lucent SVM demo
Archives of SVM
Royal Holl. SVM
SVM-light

...

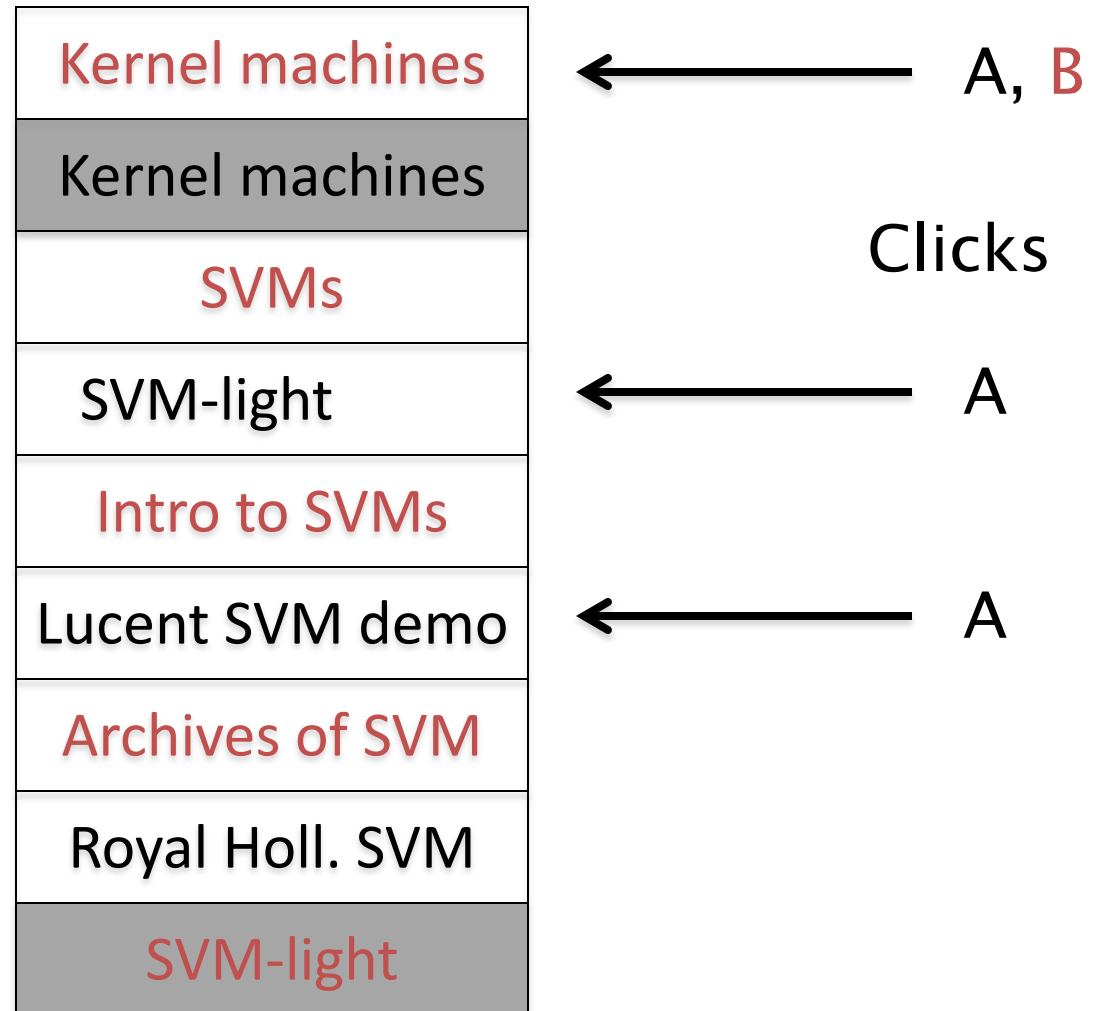
Remove duplicate results

Kernel machines
Kernel machines
SVMs
SVM-light
Intro to SVMs
Lucent SVM demo
Archives of SVM
Royal Holl. SVM
SVM-light

...

Count user clicks

Ranking A: 3
Ranking B: 1



...

Interleaved ranking

- Present interleaved ranking to users
 - Start randomly with ranking A or ranking B to even out presentation bias
- Count clicks on results from A versus results from B
- Better ranking will (on average) get more clicks

A/B testing at web search engines

- Purpose: Test a single innovation
- Prerequisite: You have a large search engine up and running.
- Have most users use old system
- Divert a small proportion of traffic (e.g., 0.1%) to an experiment to evaluate an innovation
 - Interleaved experiment
 - Full page experiment

Recap

- Benchmarks consist of
 - Document collection
 - Query set
 - Assessment methodology
- Assessment methodology can use raters, user clicks, or a combination
 - These get quantized into a *goodness measure* – Precision/NDCG etc.
 - Different engines/algorithms compared on a benchmark together with a goodness measure

User behavior

- User behavior is an intriguing source of relevance data
 - Users make (somewhat) informed choices when they interact with search engines
 - Potentially a lot of data available in search logs
- But there are significant caveats
 - User behavior data can be very noisy
 - Interpreting user behavior can be tricky
 - Spam can be a significant problem
 - Not all queries will have user behavior

Incorporating user behavior into ranking algorithm

- Incorporate user behavior features into a ranking function like BM25F
 - But requires an understanding of user behavior features so that appropriate V_j functions are used
- Incorporate user behavior features into *learned* ranking function
- Either of these ways of incorporating user behavior signals improve ranking